Department of B. Tech First Year

I[st] Sem Session 2026-27 (ODD Semester)

Assignment - 2

Subject Name: Programming Language **Subject Code: ACS 101**

Date of Assignment: 15.10.2026 **Date of Submission: 21.10.2026**

Q.No.	Questions	Marks	CO	KL
1	Differentiate between: **i).** Switch case and else if ladder **ii).** Break and default in switch **iii).** Actual and formal parameters.	6	2	K3
2	In a billing system, a certain amount of discount is given. **i).** If total purchase less than 500 no discount. **ii).** If total purchase is more than 500 and less than 1000, 10% discount. **iii).** If total purchase is more than 1000 and less than 5000, 20% discount. **iv).** If total purchase is more than 5000' 30% discount. W.A.P to calculate the total bill of a customer.	3	2	K3
3	What are the differences between while and do-while loop? Explain with the help of a suitable example.	3	2	K3
4	W.A.P to design a basic calculator that takes two operands and one operator (+, -, *, /) from the user and displays the result of the operation as per the operator entered. Also explain the use of break in the switch case.	3	2	K3
5	What are the differences between call by value and call by reference? Explain with the help of an example.	5	2	K3
6	How to call, define and declare function/function prototype, explain with the help of syntax and example.	3	2	K3
7	What is a recursive function? Explain the working of recursion with the help of an example. Also explain the role of base condition in recursion.	3	2	K3

8	What will be the output of the following? Give reasons
	main( )
	{
	int i=1;
	switch(i)
	{case1: printf("I am in case 1\n");
	printf("Thank you");
	case2: printf("I am in case 2\n");
	printf("Thank you");
	case3: printf("I am in case 3\n");
	printf("Thank you");
	default: printf("Invalid choice"); }
		2	2	K3

8	What is the difference between break and continue? Explain in detail.	2	2	K3

Signature & Name of Faculty **HoD Signature**

Date: **Date:**

CO - Course Outcomes mapped with respective question

KL - Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)

K1 – Remember, **K2 –** Understand, **K3 –** Apply, **K4 –** Analyze, **K5 –** Evaluate, **K6 -** Create

Note: Use pages to write the answers. Avoid copying assignments